Electrical Engineering Department
Final Exam
Control Systems

Name & Surname: *14.06.2023 – 09:30*

READ CAREFULLY!

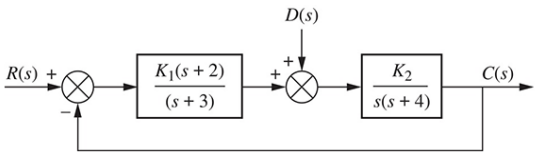
- The exam is closed Books and Notes
- Exam duration is 120 minutes
- Write this honor pledge by your own handwriting only and sign.
  "*I affirm that I will not give or receive any unauthorized help on this exam, and that all work will be my own.*"
  ……
  ……
  ……
  ……
  ……..
  Signature,

- You may write down the honor pledge on the first page only.
- Solutions without this pledge and signature will not be graded.

1- Discuss about the following items in the most appropriate and concise way (in English)
   - Does the presence of an entire row of zeros in the Routh table always mean that the system has poles on the $j\omega$ axis?
   - What are the fundamental causes for the existence of steady state error in a system? (Input signal, transfer function, characteristic equation, type, order, …) discuss about your answer.
   - Does the stability of a system changes for various input signals? why?
   - How does zero and pole placements affect stability, accuracy and performance (Max. overshoot and settling time) of the system?

Score: 20/100

2- In the given system, determine the two gains K_1 and K_2 so that both following conditions are satisfied,
   - Steady-state error due to a unit step disturbance is -0.00001
   - Steady-state error due to a unit step input is 0.002



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3- Determine the range of *K* in the following transfer function for a typical tape-drive, which leads to a stable system. Note that the closed-loop transfer function is *1+KG(s) = 0.*

$$KG(s) = \frac{K(s+6)}{s\left[(s+0.7)(s+1.2)(s^2+0.8s+6)\right]}$$

Score: 20/100

4- In a unity negative feedback system,
   a. Obtain the root locus plot if the open loop transfer function is $G(s) = \frac{K(s+1)}{s(s-1)(s^2+6s+30)}$
   b. Determine the range of *K* for stability, according to the plot.
   c. What is the root locus plot for the same system in a positive feedback architecture?

Score: 40/100

Good luck
M. Rahmanian